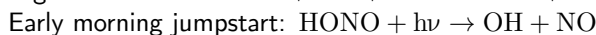
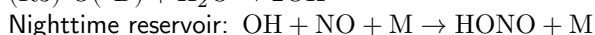
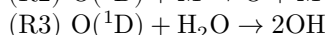
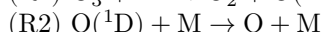


10 Oxidizing Power of the Troposphere

Tropospheric Chemistry

1. *The hydroxyl radical OH*. Hydroxyl is a short-lived free radical and by far the most effective scavenger in the troposphere. It is the main oxidant for CO, CH₄ and higher hydrocarbons, H₂S (hydrogen sulfide) and SO₂ (sulfur dioxide).
2. *The nitrate radical NO₃*. At night, this radical takes over from hydroxyl as the dominant oxidant in the atmosphere: hydroxyl is formed by photolysis and its concentration peaks during daytime while NO₃ does not survive sunlight.
3. *The oxygen atom O(³P)*. This excited state of the oxygen atom has the ability to oxidize unsaturated hydrocarbons and other gases containing a double bond such as CS₂ and COS in the upper troposphere.
4. *Peroxy and hydroperoxy radicals HO₂ and RO₂ (where R is an alkyl)*. HO₂ and RO₂ are very much intertwined with hydroxyl in the oxidation cycle. They are not as efficient as hydroxyl, but react with themselves to form H₂O₂, an important oxidant in cloud droplets.
5. *Hydrogen peroxide H₂O₂*. This strong acid reacts very efficiently in cloud droplets and oxidizes a number of trace gases, in particular sulfur dioxide. Highly soluble, it also accounts for a large part of the excess acidity in rain.

Production of OH



Production rate of [OH]:

$$P_{\text{OH}} = 2k_3[\text{O}(^1\text{D})][\text{H}_2\text{O}] = 2k_1k_3[\text{O}_3][\text{H}_2\text{O}]/(k_2[\text{M}] + k_3[\text{H}_2\text{O}]) \approx 2k_1k_3[\text{O}_3][\text{H}_2\text{O}]/k_2[\text{M}] \quad (\text{since } k_2[\text{M}] \gg k_3[\text{H}_2\text{O}])$$

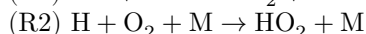
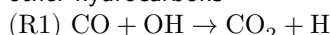
$$P_{\text{OH}} \approx \frac{2k_1k_3}{k_2[\text{M}]}[\text{O}_3][\text{H}_2\text{O}]$$

Methyl chloroform (CH₃CCl₃) is removed from the atmosphere almost solely by reaction with OH, and hence it is possible to infer the global mean OH level. Global mean [OH] $\approx 1.5 \times 10^6$ molecules cm⁻³.

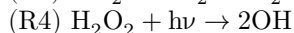
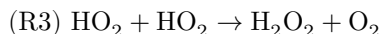
$$\frac{dN}{dt} = P - L_{\text{trop}} - L_{\text{strat}} - L_{\text{ocean}} \approx P - L_{\text{trop}}, L_{\text{trop}} = \int_{\text{trop}} k(T)Cn_a[\text{OH}] dV_{\text{trop}}, [\text{OH}] \approx \frac{P - dN/dt}{C \int_{\text{trop}} k(T)n_a dV_{\text{trop}}}$$

Oxidation Mechanism of CO

Source of CO: fossil fuel combustion, biomass burning, oxidation of CH₄ and other hydrocarbons

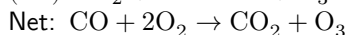
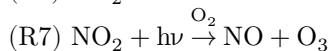


● Low NO_x condition



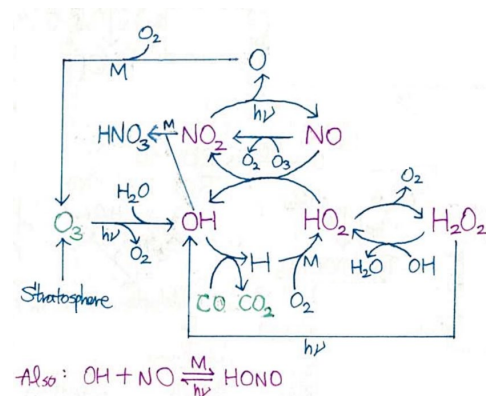
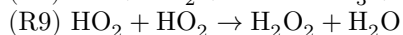
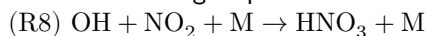
No generation of O₃.

● High NO_x condition



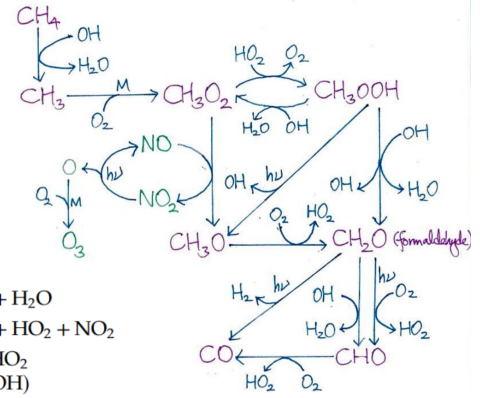
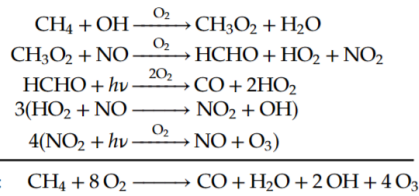
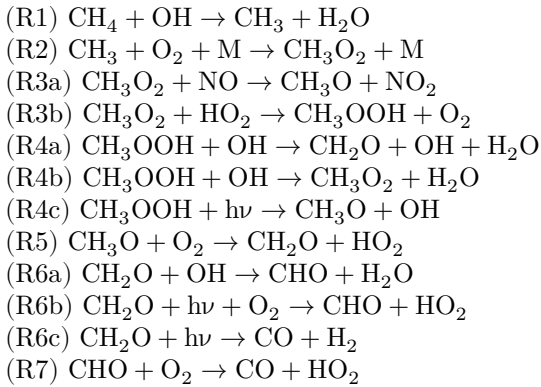
No net consumption of OH and HO₂, O₃ is formed during conversion of NO₂ to NO.

● Terminating step



Oxidation Mechanism of Methane

- Source of CH₄: wetlands, rice paddies, livestock, natural gas extraction, termites



- Under high NO_x environment, (R3b), (R6b) are preferred. Oxidation of CO can produce one more molecule of O₃, but the maximum yield is not realized in the atmosphere due to the neglected competing reactions.
- In clean atmosphere, ~70% of OH reacts with CO, ~30% with CH₄.

NO_x and NO_y Families

- Source of NO_x: fossil fuel combustion, biomass burning, soil microorganism, lightning, NH₃ oxidation.
- NO_x sources are mainly anthropogenic and mostly emitted as NO.
- NO_y = NO + NO₂ + HNO₃ + HONO + NO₃ + N₂O₅ + RONO + RONO₂ + ROONO₂
- Rapid cycling of NO and NO₂
 - (R1) $\text{HO}_2 + \text{NO} \rightarrow \text{OH} + \text{NO}_2$
 - (R2) $\text{CH}_3\text{O}_2 + \text{NO} \rightarrow \text{CH}_3\text{O} + \text{NO}_2$
 - (R3) $\text{O}_3 + \text{NO} \rightarrow \text{NO}_2 + \text{O}_2$
 - (R4) $\text{NO}_2 + h\nu \rightarrow \text{NO} + \text{O}$
 - (R5) $\text{O} + \text{O}_2 + \text{M} \rightarrow \text{O}_3 + \text{M}$

$$\frac{d[\text{NO}_2]}{dt} = -J_{\text{NO}_2}[\text{NO}_2] + k_3[\text{O}_3][\text{NO}], \quad \frac{d[\text{O}]}{dt} = J_{\text{NO}_2}[\text{NO}_2] - k_5[\text{O}][\text{O}_2][\text{M}]$$

PSSA of [O]: $J_{\text{NO}_2}[\text{NO}_2] = k_5[\text{O}][\text{O}_2][\text{M}]$

$$[\text{O}] = \frac{J_{\text{NO}_2}[\text{NO}_2]}{k_5[\text{O}_2][\text{M}]}, \quad [\text{O}_3] = \frac{J_{\text{NO}_2}[\text{NO}_2]}{k_3[\text{NO}]}$$

A typical 12-h daytime average ratio is [NO]/[NO₂] is approximately 0.1.

- Sink of NO_x (Daytime)
 - (R7) $\text{OH} + \text{NO}_2 + \text{M} \rightarrow \text{HNO}_3 + \text{M}$
 - NO₃ radicals photolyze rapidly in daytime
 - (R8) $\text{NO}_3 + h\nu \rightarrow \text{NO} + \text{O}_2$ ($\lambda < 700$ nm)
 - (R9) $\text{NO}_3 + h\nu \rightarrow \text{NO}_2 + \text{O}$ ($\lambda < 580$ nm)
 - (R10) $\text{NO}_3 + \text{NO} \rightarrow 2\text{NO}_2$

A few ppt in daytime.

- Sinks of NO_x (Nighttime)
 - (R11) $\text{NO}_2 + \text{O}_3 \rightarrow \text{NO}_3 + \text{O}_2$
 - (R12) $\text{NO}_2 + \text{NO}_3 + \text{M} \rightarrow \text{N}_2\text{O}_5 + \text{M}$
 - (R13) $\text{N}_2\text{O}_5 + \text{M} \rightarrow \text{NO}_2 + \text{NO}_3 + \text{M}$
 - (R14) $\text{N}_2\text{O}_5 + \text{H}_2\text{O(s)} \rightarrow 2\text{HNO}_3$
 - HNO₃ can reach more than 100 ppt in nighttime.
 - Equilibrium of (R12) and (R13) can be established in a few minutes.

Ozone Budget of the Troposphere

O₃ can be transported from the stratosphere or undergo dry deposition.

In situ chemical production of O₃ results primarily from reactions of peroxy radicals with NO; in situ chemical loss of O₃ results mainly from O(¹D) + H₂O, O₃ + HO₂ and O₃ + OH.

11 Tropospheric Ozone Production

Primary Pollutants

- Released directly from sources.
- Examples: CO, SO₂, NO_x

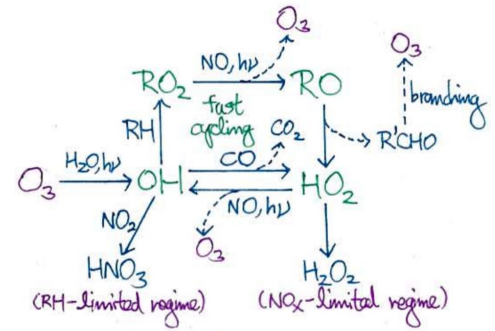
Secondary Pollutants

- Formed through chemical reactions from primary pollutants in the air.
- Example: O₃

Good ozone protects Earth from UV radiation, while bad ozone causes smog, breathing issues, and environmental damage. Smog = aerosols + invisible toxic gases such as ozone + SO₂ + CO.

Formation of Ozone

- NO₂ is capable of absorbing visible light to produce O.
 - (R1) $\text{NO}_2 + h\nu \rightarrow \text{NO} + \text{O}$ ($\lambda < 400 \text{ nm}$)
 - (R2) $\text{O} + \text{O}_2 + \text{M} \rightarrow \text{O}_3 + \text{M}$
- Null Cycle
 - (R3a) $\text{NO} + \text{O}_3 \rightarrow \text{NO}_2 + \text{O}_2$
- If NO is oxidized by other oxidants, O₃ can accumulate, followed by (R1).
 - (R3b) $\text{NO} + \text{HO}_2 \rightarrow \text{NO}_2 + \text{OH}$
 - (R3c) $\text{NO} + \text{RO}_2 \rightarrow \text{NO}_2 + \text{RO}$
- Production under low NO_x conditions
 - (R4) $\text{HO}_2 + \text{HO}_2 \rightarrow \text{H}_2\text{O}_2 + \text{O}_2$
 - (R5) $\text{NO}_2 + \text{OH} + \text{M} \rightarrow \text{HNO}_3 + \text{M}$
- Production under high NO_x conditions
 - (R6) $\text{RH} + \text{OH} \xrightarrow{\text{O}_2} \text{RO}_2 + \text{H}_2\text{O}$
 - (R7) $\text{RO} + \text{O}_2 \rightarrow \text{R}'\text{CHO} + \text{HO}_2$
- Rate Limiting Steps
 - (R8) $\text{O}(^1\text{D}) + \text{H}_2\text{O} \rightarrow 2\text{OH}$
 - (R9) $\text{O}_3 + \text{HO}_2 \rightarrow \text{OH} + 2\text{O}_2$
 - (R10) $\text{O}_3 + \text{OH} \rightarrow \text{HO}_2 + \text{O}_2$



Assume the chain propagation is efficient so that $k_{\text{RH}+\text{OH}} = k_{\text{RO}_2+\text{NO}} = k_{\text{RO}+\text{O}_2} = k_{\text{HO}_2+\text{NO}}$

PSSA of NO_x: $[\text{O}_3] = \frac{J_{\text{NO}_2}[\text{NO}_2]}{k_{\text{NO}+\text{O}_3}[\text{O}_3]}$, $P_{\text{O}_3} = (k_{\text{HO}_2+\text{NO}}[\text{HO}_2] + k_{\text{RO}_2+\text{NO}}[\text{RO}_2])[\text{NO}] = 2k_{\text{HO}_2+\text{NO}}[\text{HO}_2][\text{NO}]$

PSSA of OH: $[\text{OH}] = \frac{k_{\text{HO}_2+\text{NO}}[\text{HO}_2][\text{NO}]}{k_{\text{RH}+\text{OH}}[\text{RH}]}$,

PSSA of HO_x: $P_{\text{HO}_x} = 2k_{\text{HO}_2+\text{HO}_2}[\text{HO}_2]^2 + k_{\text{NO}_2+\text{OH}+\text{M}}[\text{NO}_2][\text{OH}][\text{M}]$

High NO_x: $P_{\text{HO}_x} \approx k_{\text{NO}_2+\text{OH}+\text{M}}[\text{NO}_2][\text{OH}][\text{M}]$

$$[\text{OH}] = \frac{P_{\text{HO}_x}}{k_{\text{NO}_2+\text{OH}+\text{M}}[\text{NO}_2][\text{M}]} = \frac{k_{\text{HO}_2+\text{NO}}[\text{HO}_2][\text{NO}]}{k_{\text{RH}+\text{OH}}[\text{RH}]} \Rightarrow [\text{HO}_2] = \frac{P_{\text{HO}_x} k_{\text{RH}+\text{OH}}[\text{RH}]}{k_{\text{HO}_2+\text{NO}} k_{\text{NO}_2+\text{OH}}[\text{NO}][\text{NO}_2][\text{M}]}$$

$P_{\text{O}_3} = \frac{2k_{\text{RH}+\text{OH}} P_{\text{HO}_x} [\text{RH}]}{k_{\text{NO}_2+\text{OH}}[\text{NO}_2][\text{M}]}$, which increases linearly with hydrocarbon concentrations but varies inversely with NO_x

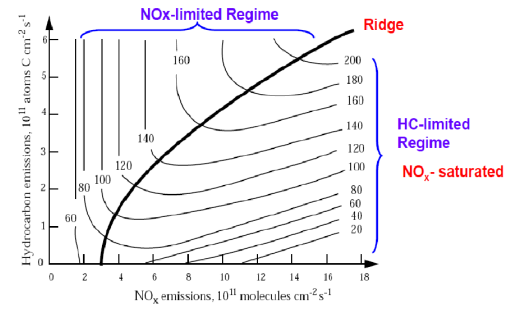
concentrations. Hence, O₃ formation is hydrocarbon-limited.

Low NO_x: $P_{\text{HO}_x} \approx 2k_{\text{HO}_2+\text{HO}_2}[\text{HO}_2]^2 \Rightarrow [\text{HO}_2] = \left(\frac{P_{\text{HO}_x}}{2k_{\text{HO}_2+\text{HO}_2}} \right)^{1/2}$, $P_{\text{O}_3} = 2k_{\text{HO}_2+\text{NO}} \left(\frac{P_{\text{HO}_x}}{2k_{\text{HO}_2+\text{HO}_2}} \right)^{1/2} [\text{NO}]$

O₃ production varies linearly with [NO] but is independent of hydrocarbons. Hence, O₃ formation is NO_x-limited.

Ozone Isopleth Diagram

- In the NO_x-limited regime, hydrocarbon emission controls are of no benefit for reducing O₃ formation (e.g., Boston)
- In the hydrocarbon-limited regime, NO_x emission controls cause an increase in O₃.
- The ridges separate the NO_x and hydrocarbon-limited regimes.



Ozone Control Strategies

Clean air act: reduce VOCs since they never increase O₃ but is not the case for NO_x.

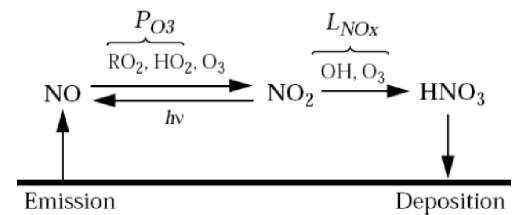
In the 1970s, ozone production was generally in HC-limited regime. Hence, VOC emission control is ineffective for O₃ reduction except for LA and New York. Initial models significantly underestimated mobile source VOC emissions and biogenic VOCs.

In the 1990s, discovery found that an area in the RH-limited regime has switched to NO_x-limited, which has led to the Revised Clean Air Act of 1990 for simultaneous reductions of both VOCs and NO_x.

Ozone Production Efficiency (OPE)

$$\varepsilon = \frac{P_{O_3}}{L_{NO_x}} = \frac{2k_{HO_2+NO}[HO_2][NO]}{k_{NO_2+OH+M}[NO_2][OH]},$$

which is the number of O₃ molecules produced per molecule of NO_x consumed. ε is the chain length when NO_x is viewed as the propagating agent in the O₃ production. A NO_x molecule emitted to the atmosphere undergoes a number ε of peroxy + NO reactions, producing O₃, before being converted to HNO₃ which is removed mainly by deposition.



Assuming efficient HO_x cycle, i.e. PSSA for [OH]: $[OH] = \frac{k_{HO_2+NO}[HO_2][NO]}{k_{RH+OH}[RH]}$

$$\varepsilon = \frac{P_{O_3}}{L_{NO_x}} = \frac{2k_{RH+OH}[RH]}{k_{NO_2+OH+M}[NO_2]}$$

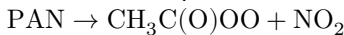
- RH concentration effect: increasing RH concentrations causes OH to decrease ($OH + RH \rightarrow$ increase the lifetime of NO_x $\rightarrow$ allow more O₃ production per molecule of NO_x emitted).
- NO_x concentration effect: increasing NO_x causes OH to increase ($HO_2 + NO \rightarrow OH + NO_2$) and favors the termination process ($OH + NO_2 \rightarrow HNO_3$); the negative dependence of ε on [NO_x] dampens the O₃ decrease to be expected from a given reduction of NO_x emissions (For example, 50% reduction of NO_x emissions $\rightarrow$ decrease summertime O₃ concentrations by ~15%.)

Peroxyacetyl Nitrate (PAN)

- PAN: CH₃C(O)OONO₂
- Formation of PAN
 - (R1) CH₃CHO + OH $\rightarrow$ CH₃CO + H₂O
 - (R2) CH₃CO + O₂ + M $\rightarrow$ CH₃C(O)OO + M
 - (R3) CH₃C(O)OO + NO₂ + M $\rightarrow$ CH₃C(O)OONO₂ + M
- Properties of PAN

Sparing soluble in water and is not efficiently removed by wet deposition.

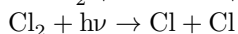
Thermal decomposition is strongly dependent on temperature.



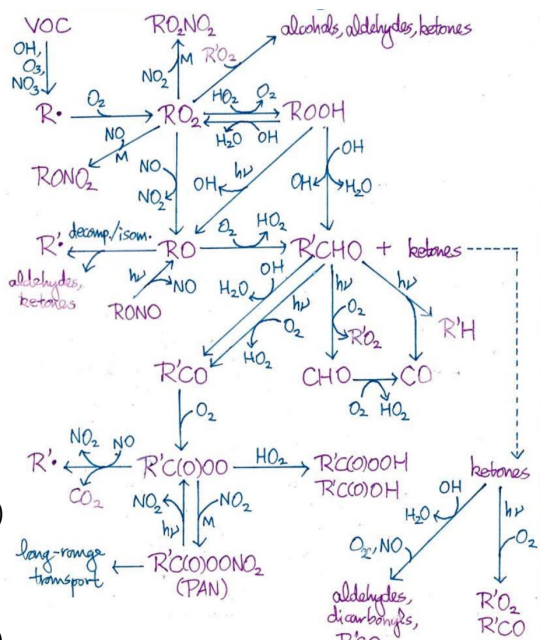
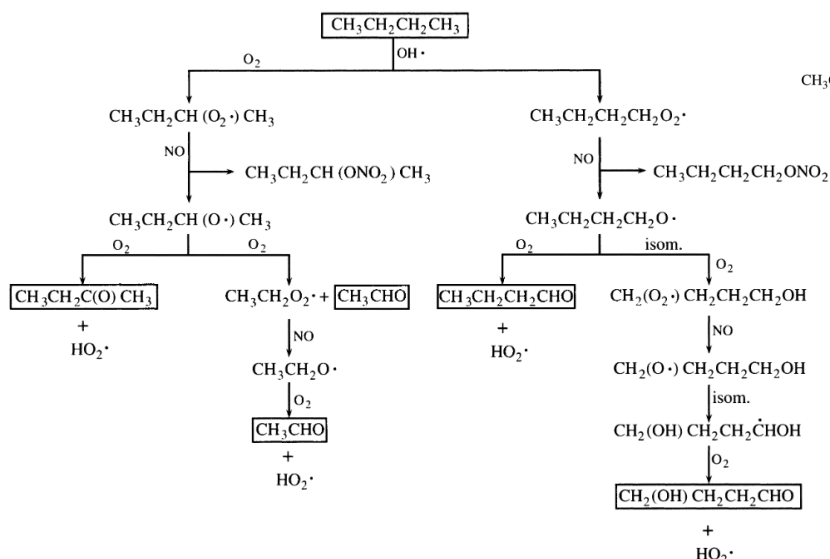
At 295K, $\tau_{PAN} = 1$ hr; At 250K, τ_{PAN} can be up to several months and transported over long distances to release NO_x far from the source. PAN is the most abundant organic nitrates that contribute to the far-region NO_x release.

Atmospheric Regions

Sea Salt Chemistry

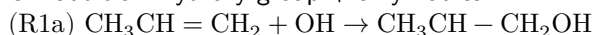


O₂ abstraction, isomerization (dominates if possible), fragmentation

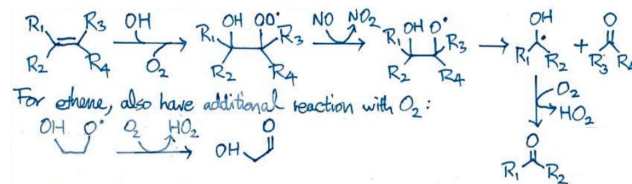


Alkene Oxidation Mechanism

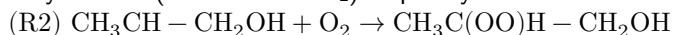
- OH addition: hydroxy group + alkyl radical



For dienes with conjugated double bonds, OH radical addition to the $\text{C}=\text{C}-\text{C}=\text{C}$ system occurs preferentially at positions 1 and/or 4.



- Alkyl radical (reaction with O_2) to peroxy radical



- Peroxy radical to alkoyl radical or (stable) nitrate

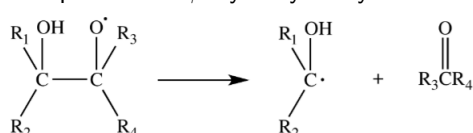
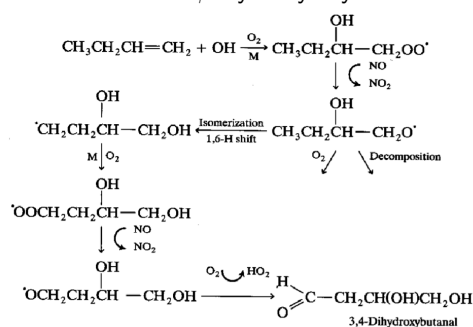


- Alkoxy radical: reaction with O_2 , decomposition, isomerization



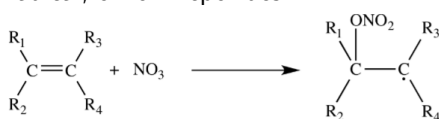
- Isomerization of β -hydroxyalkyl

Decomposition of β -hydroxyalkoxy radicals ($\geq \text{C}_3$ alkenes competitive)

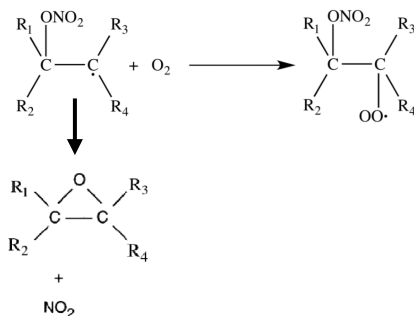
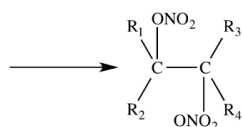
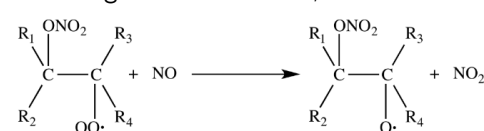


- NO_3 reaction with biogenic alkenes at night are very important

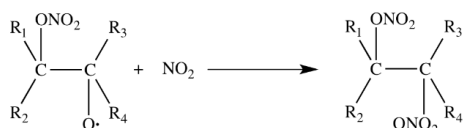
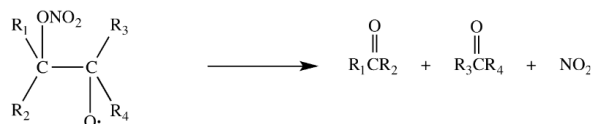
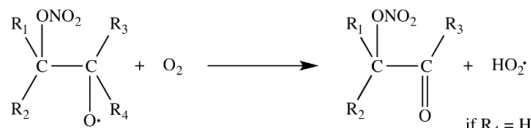
H-atom abstraction is relatively insignificant, then followed by rapid O_2 addition to produce the β -nitroalkyl peroxy radical, or form epoxides.



Under high- NO_x conditions,



Further reactions of β -nitroalkoxy radicals include



where last reaction is expected to be minor.

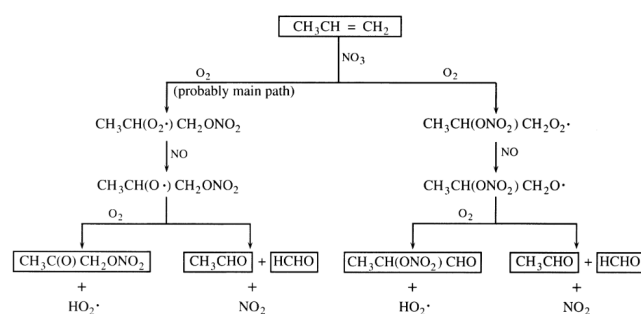
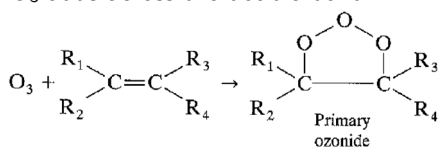


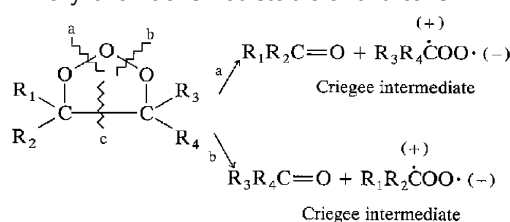
FIGURE 6.20 Propene- NO_3 reaction mechanism.

- O_3 reaction

O_3 adds across the double bond

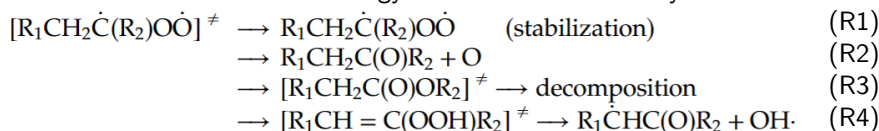


Primary ozonide is not stable and breaks



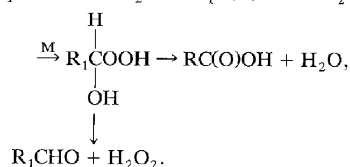
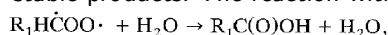
- Fate of excited Criegee intermediates (R_1R_2COO)*

Broken bonds contain excess energy and can be stabilized by collision or decomposition.



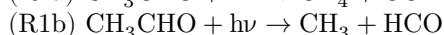
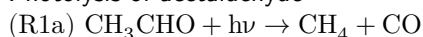
At atmospheric pressure, O atoms are not formed in any appreciable amount, so the second path above can generally be neglected.

The stabilized Criegee intermediates can react with H_2O , SO_2 , NO , NO_2 , CO , aldehydes, and ketones and lead to stable products. The reaction with H_2O dominates.



Aldehyde Oxidation Mechanisms

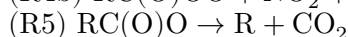
- Photolysis of acetaldehyde



- H-atom abstraction by OH and further oxidation by O_2

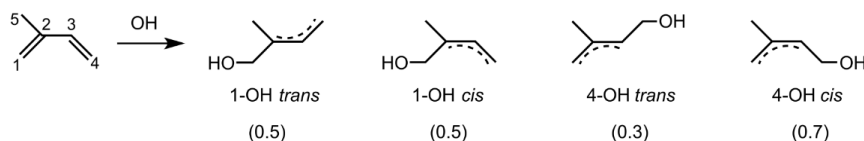


- Reaction with NO , NO_2



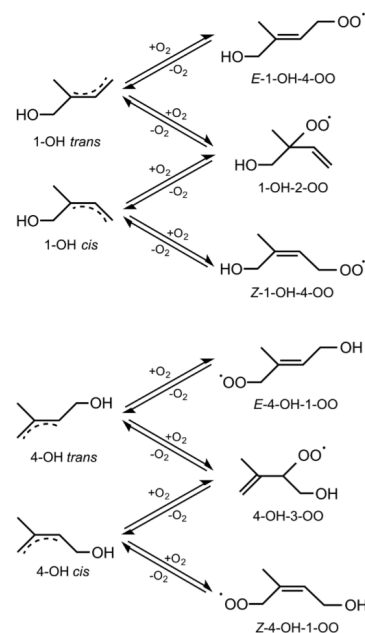
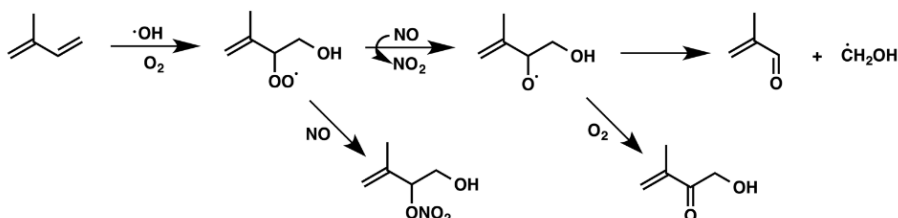
Isoprene Chemistry

- Addition of OH

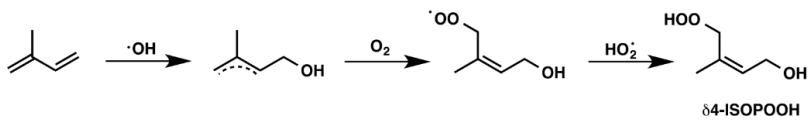
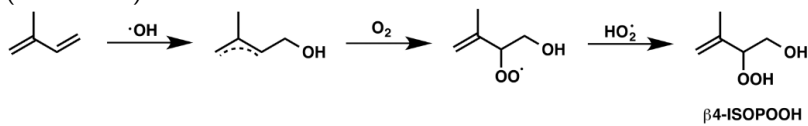


- O_2 adds immediately following addition of OH to the isoprene molecule, with an overall reaction rate constant of $2.3 \times 10^{-12} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$, leading to the formation of six distinct peroxy radicals. Whereas reverse reaction (O_2 loss from RO_2) is negligible for most atmospheric peroxy radicals, the reverse reactions for the six major peroxy radicals formed in the OH oxidation of isoprene are not negligible.

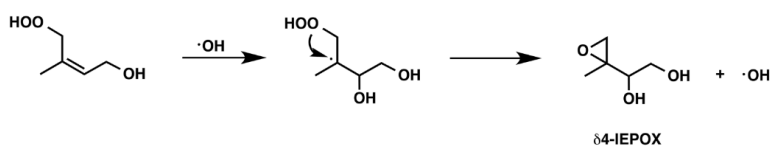
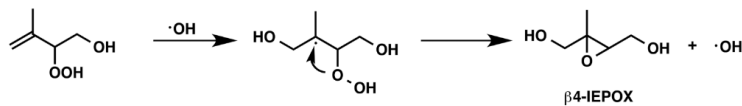
- Fate of the isoprene peroxy radicals depends on the relative concentrations of NO and HO_2 . The principal product is the unsaturated aldehyde, methacrolein (MACR).



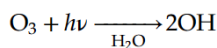
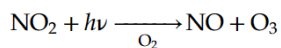
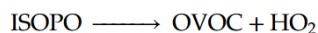
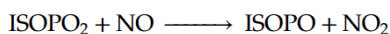
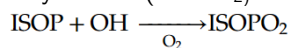
- Since the atmospheric concentration of HO₂ generally exceeds that of other RO₂ species, the principal reaction of the isoprene peroxy radical under sufficiently low NO conditions is with HO₂, yielding isoprene hydroxyhydroperoxides (ISOPOOH):



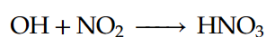
- ISOPOOH is further oxidized by OH under low-NO conditions to form isoprene epoxydiols (IEPOX), β -IEPOX isomers account for >97% IEPOX formed from isoprene:



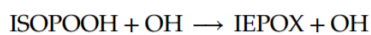
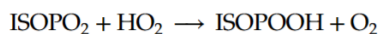
- IEPOX is expected to undergo hundreds of collisions with aerosol surfaces before reacting with OH; as a result, it plays a key role in the formation of secondary organic aerosol from isoprene.
- The OH oxidation of isoprene under conditions in which the concentration of NO is sufficiently high that the isoprene peroxy radical (ISOPO₂) reacts exclusively with NO can be represented by the general mechanism



- The radical chain is terminated by formation of nitric acid and hydrogen peroxide:



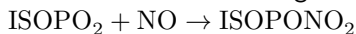
- When NO levels are sufficiently low that ISOPO₂ reacts preferentially with HO₂ rather than NO:



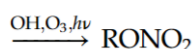
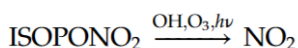
- Unimolecular isomerization of ISOPO₂ can be competitive with reaction



- Modulation of NO_x through formation of isoprene nitrates, slows down O₃ formation



- Release of NO_x from ISOPONO₂



13 Atmospheric Aerosols

Aerosol

- stable suspension of solid or liquid particles in a gas.
- particles are liquid or solid object 0.001–100 μm in diameter, suspended by Brownian motion or convection for minutes to years.
- PM are referred as the collection of all particles suspended in a given volume of air ($\mu\text{g}/\text{m}^3$).
- Primary particles: Direct emission (combustion, dust, sea salt, debris).
- Secondary particles: Formed from gases (nucleation, condensation).

Optical Properties and Terminologies

- **Scattering:** Light redirected without absorption.
- **Reflection:** Abrupt change in direction of a light beam at an interface between two dissimilar media so that the light beam returns to the medium from which it originated.
- **Refraction:** Bending of light as it passes from one medium into another.
- **Diffraction:** Deviation of light from a straight-line propagation due to wave obstruction phenomena.
- **Rayleigh scattering** (small particles): $S \propto \frac{r^6}{\lambda^4}$
blue light scattered more $\rightarrow$ sky is blue.
- **Criteria for reduced visibility** in Hong Kong: $> 8 \text{ km}$, $\text{RH} < 95\%$, no fog/mist/precipitation. More aerosols $\rightarrow$ more scattering $\rightarrow$ lower visibility.

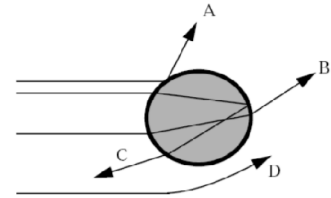


Figure Scattering of a radiation beam: processes of reflection (A), refraction (B), refraction and internal reflection (C), and diffraction (D).

Effects of Aerosols on Climate

- Direct effect: aerosol scatter light $\rightarrow$ increase the Earth's albedo $\rightarrow$ cooling.
- Indirect effect: cloud form by condensation on CCN with suppressed precipitation.

Physical Processes of Particle Production and Growth

- **Nucleation:** Homomolecular (one species) or heteromolecular (binary).
- **Condensation/Evaporation:** Gas molecules captured/released at the surface of existing particles.
Continuum regime ($D_p > 0.2 \mu\text{m}$):
Total flow of A(g) (moles time^{-1}) toward the particle $J_c = 4\pi R_p D_g (c_\infty - c_s)$, R_p is the particle radius, D_g is the diffusivity of gas A, c_∞ is the concentration of A far from the particle and c_s is the vapor phase concentration of A at the particle surface.
Free molecular ($D_p < 0.01 \mu\text{m}$):
Total flow of A(g) (moles time^{-1}) toward the particle $J_k = \pi R_p^2 \bar{c}_A \alpha (c_\infty - c_s)$, $\bar{c}_A = \left(\frac{8kT}{\pi m_A}\right)^{1/2}$ is the mean speed of the moles and α is the molecular accommodation coefficient, i.e. probability of A to stick on particle.
- **Coagulation:** Particles collide and stick. Rate $= K_{12} N_1 N_2$ depends on the diameter of the large particle, diffusion rates of the smaller particle and the concentration of the particles. Smallest for same-size particles.
- **Deposition:**
Dry Deposition: size-dependent.
Wet Deposition: rainout (in-cloud, removed when cloud droplets form rain drops) + washout (below-cloud, falling hydrometers capturing particles by impaction).

Aerosol Particle Size and Distribution Modeling

- **Aerodynamic diameter** (D_a): Diameter of a unit-density sphere with same terminal velocity.

$$D_a = D_g \sqrt{\frac{\rho_p}{\rho_0 \cdot \chi}}$$

- **Particle size matters** for:
 - (i) toxicity & human health (respiratory deposition), as it determines where in the respiratory tract the particle deposits;
 - (ii) light scattering & visibility, the size range that best scatters visible light is $\sim 0.1\text{--}1 \mu\text{m}$, which causes the greatest haze and visibility impairment;
 - (iii) atmospheric lifetime & transport, accumulation mode particles has the longest atmospheric lifetime and thus the greatest impact on regional/global air quality and climate;
 - (iv) surface reactions & chemical activity, as smaller particles have larger surface-to-volume ratio, providing more sites for heterogeneous reactions.

- **Modes**

Coarse mode ($\geq 2.5 \mu\text{m}$): mechanical processes, dominate mass.

Fine mode ($\leq 2.5 \mu\text{m}$)

Aitken ($\leq 0.1 \mu\text{m}$)

Accumulation (0.1–2.5 μm): chemical processes, dominate number.

- **Lognormal distribution is typical**

$$n_N(\log D_p) = \sum \frac{N_i}{\sqrt{2\pi} \log \sigma_i} \exp \left(-\frac{(\log D_p - \log \bar{D}_{pi})^2}{2 \log^2 \sigma_i} \right)$$

- Numbers of particles ranging from 100–300 cm^{-3} (pristine remote areas) to 10^8 – 10^9cm^{-3} (polluted urban areas).

Cloud Formation – Köhler Theory

- Kelvin effect: Smaller droplets have higher equilibrium vapor pressure (curvature).
- Raoult's law: Solution lowers equilibrium vapor pressure.
- Köhler equation:

$$\ln \left(\frac{p_w(D_p)}{p^o} \right) = \frac{A_w}{D_p} - \frac{B_s}{D_p^3}$$

- Larger or more soluble particles activate more easily.

Chemical Composition

- **Inorganics:** Sulfate, nitrate, ammonium, crustal, sea salt, H^+ , water.
- **Organics:** Elemental carbon (EC), organic carbon (OC) – thousands of compounds, mostly unidentified.
- **Difficulty:** $\sim \text{pg m}^{-3}$ level concentration, unstable species

Formation of Sulfate-Nitrate-Ammonium

- **Gas-phase:**
 $\text{SO}_2 + \text{OH} \rightarrow \text{H}_2\text{SO}_4 \rightarrow \text{neutralized by } \text{NH}_3 \rightarrow (\text{NH}_4)_2\text{SO}_4$
 $\text{NO}_2 + \text{OH} \rightarrow \text{HNO}_3 \rightarrow \text{with excess } \text{NH}_3 \rightarrow \text{NH}_4\text{NO}_3$
- **Aerosol-phase:**
 Sulfate always aqueous, consumes NH_3 first.
 Nitrate only taken up if excess NH_3 remains after sulfate neutralization.
 Low RH $\rightarrow$ solid NH_4NO_3 .

Gas-Aerosol Partitioning (Organic Aerosol)

- Absorption: Molecules of one substance are taken up into the bulk volume (interior) of another solid or liquid. Uniform distribution throughout the absorbing material.
- Adsorption: Molecules of a substance (adsorbate) adhere to the surface of a solid or liquid (adsorbent). Accumulation at the interface, not penetration.
- Equilibrium partitioning: Raoult's law basis.
- Absorption model

$$\xi_i = \frac{1}{1 + C_i^*/C_{OA}}$$

where ξ_i = fraction of compound i in condensed phase, C_i^* = effective saturation concentration, C_{OA} = total organic aerosol mass.

- Langmuir adsorption isotherm model

Assumptions:

1. all adsorption sites on the surface are equal.
2. no horizontal interactions among adsorbed molecules.
3. heat of adsorption for all molecules is the same.

Langmuir equation:

$$\theta = \frac{KP}{1 + KP}$$

where θ is fraction of adsorption sites on a surface that are occupied by adsorbed molecules, K the equilibrium constant, and P the gas pressure.

- Two-product model:

Oxidation of hydrocarbon $\rightarrow$ two semivolatile products with different volatilities.

$$\text{Yield } Y = M \sum \frac{\alpha_i K_{p,i}}{1 + MK_{p,i}}$$

Only 4 parameters needed ($\alpha_1, \alpha_2, K_1, K_2$).

14 Aqueous Phase Chemistry

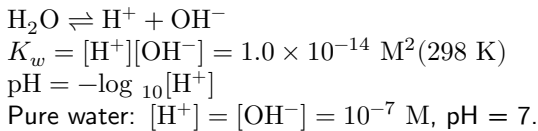
Importance of Aqueous Phase Chemistry

- Clouds cover ~60% of Earth's surface, which act as a major reactor medium.
- Aqueous chemistry occurs in clouds, fog, rain, and deliquesced aerosols.
- Non-ideal solutions (high ionic strength) in aerosols are vastly different from dilute lab conditions.

Henry's Law

- $A(g) + H_2O \rightleftharpoons A \cdot H_2O$, $H_A = \frac{[A \cdot H_2O]}{p_A}$ (M/atm), $[A \cdot H_2O]$ is in g (L of air)⁻¹
- Temperature dependence (van't Hoff equation): $H(T) = H_{298} \exp \left(-\frac{\Delta H}{R} \left(\frac{1}{T} - \frac{1}{298} \right) \right)$

Water Autoionization



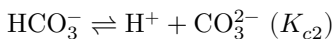
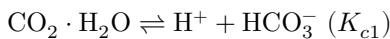
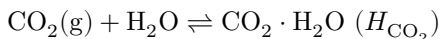
Liquid Water Content (LWC) & Partitioning

- Liquid water amount affects the partitioning of species between the gas and the aqueous phase (especially for the very soluble species). Liquid water content is often expressed for convenience as liquid water mixing ratio w_L in (volume water/volume air) and is related to L by $w_L = 10^{-6} L$ (g · m⁻³)
- Distribution factor:

$$f_A = \frac{[A(aq)]}{[A(g)]} = 10^{-6} H_A R T L = H_A R T w_L (L = \text{LWC in m}^3 \text{ water/m}^3 \text{ air})$$

- Typical $L \sim 10^{-6} \rightarrow f_A = 1$ when $H_A \sim 4 \times 10^4 \text{ M/atm}$.
- Highly soluble species (HNO_3 , H_2O_2) $\rightarrow$ mostly in aqueous phase.
- Low solubility (CO , O_3) $\rightarrow$ mostly gas phase.

CO₂ – Water Equilibrium



$$[CO_2 \cdot H_2O] = H_{CO_2} p_{CO_2}, [HCO_3^-] = \frac{H_{CO_2} K_{c1} p_{CO_2}}{[H^+]}, [CO_3^{2-}] = \frac{H_{CO_2} K_{c1} K_{c2} p_{CO_2}}{[H^+]^2}$$

Total dissolved CO₂

$$[CO_2^T] = H_{CO_2} p_{CO_2} \left(1 + \frac{K_{c1}}{[H^+]} + \frac{K_{c1} K_{c2}}{[H^+]^2} \right)$$

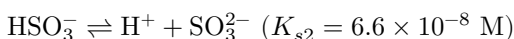
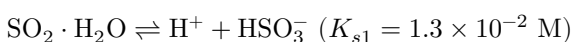
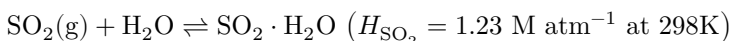
Electroneutrality (pure CO₂ – water system):

$$[H^+] = [OH^-] + [HCO_3^-] + 2[CO_3^{2-}], [H^+]^3 - (K_w + H_{CO_2} K_{c1} p_{CO_2}) [H^+] - 2H_{CO_2} K_{c1} K_{c2} p_{CO_2} = 0$$

Result at 350 ppm CO₂, 298 K: $pH = 5.6$ ("pure rainwater", slightly acidic).

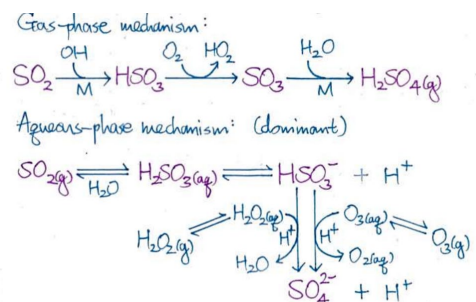
SO₂ – Water Equilibrium (S(IV))

Sources: combustion of S-containing fuels (especially coal), smelting of S-containing gas, oxidation of reduced S-gases from volcanoes and microbial activities.



$$[SO_2 \cdot H_2O] = H_{SO_2} p_{SO_2}, [HSO_3^-] = \frac{H_{SO_2} K_{s1} p_{SO_2}}{[H^+]},$$

$$[SO_3^{2-}] = \frac{H_{SO_2} K_{s1} K_{s2} p_{SO_2}}{[H^+]^2}, \text{ Total S(IV): } [S(IV)] = H_{SO_2} p_{SO_2} \left(1 + \frac{K_{s1}}{[H^+]} + \frac{K_{s1} K_{s2}}{[H^+]^2} \right)$$



Mole fractions of S(IV) species (pH dependent, independent of p_{SO_2}):

$$x_{\text{SO}_2 \cdot \text{H}_2\text{O}} = \left(1 + \frac{K_{s1}}{[\text{H}^+]} + \frac{K_{s1}K_{s2}}{[\text{H}^+]^2}\right)^{-1}, x_{\text{HSO}_3^-} = \left(1 + \frac{[\text{H}^+]}{K_{s1}} + \frac{K_{s2}}{[\text{H}^+]}\right)^{-1}, x_{\text{SO}_3^{2-}} = \left(1 + \frac{[\text{H}^+]}{K_{s2}} + \frac{[\text{H}^+]^2}{K_{s1}K_{s2}}\right)^{-1}$$

Electroneutrality condition: $[\text{H}^+] = [\text{OH}^-] + [\text{HSO}_3^-] + 2[\text{SO}_3^{2-}]$

Substitute the expressions for $[\text{OH}^-] = K_w/[\text{H}^+]$, and the two S(IV) anions:

$$[\text{H}^+] = \frac{K_w}{[\text{H}^+]} + \frac{K_{a1}H_{\text{SO}_2}p_{\text{SO}_2}}{[\text{H}^+]} + \frac{2K_{a1}K_{a2}H_{\text{SO}_2}p_{\text{SO}_2}}{[\text{H}^+]^2}$$

$$[\text{H}^+]^3 - (K_w + K_{a1}H_{\text{SO}_2}p_{\text{SO}_2})[\text{H}^+] - 2K_{a1}K_{a2}H_{\text{SO}_2}p_{\text{SO}_2} = 0$$

If sulfate (SO_4^{2-}) is also present (e.g., from oxidation or from sea salt). The electroneutrality equation becomes:

$$[\text{H}^+] = [\text{OH}^-] + [\text{HSO}_3^-] + 2[\text{SO}_3^{2-}] + 2[\text{SO}_4^{2-}]$$

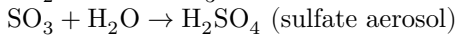
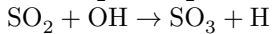
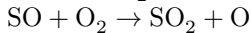
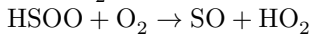
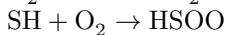
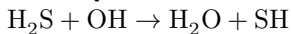
Substituting and rearranging leads to:

$$0 = [\text{H}^+]^3 - (K_w + K_{a1}H_{\text{SO}_2}p_{\text{SO}_2})[\text{H}^+] - 2K_{a1}K_{a2}H_{\text{SO}_2}p_{\text{SO}_2} - 2[\text{SO}_4^{2-}][\text{H}^+]^2$$

This equation must be solved numerically because $[\text{SO}_4^{2-}]$ itself may depend on pH (if it comes from a weak acid, e.g., HSO_4^- dissociation).

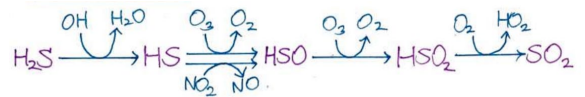
Key pH ranges: pH < 2: mostly $\text{SO}_2 \cdot \text{H}_2\text{O}$, pH 3–6: mostly HSO_3^- , pH > 7: mostly SO_3^{2-}

Atmospheric Oxidation of H_2S

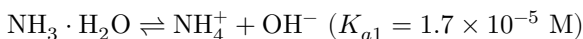
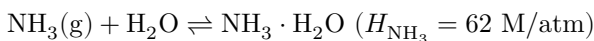


Due to the relatively slow rate constant for the $\text{OH} + \text{H}_2\text{S}$ reaction, H_2S has an atmospheric lifetime of approximately 1-2 days. Lifetime is long enough to allow for regional and even some intercontinental transport, acting as a key precursor of SO_2 .

Aqueous-phase oxidation of H_2S by O_2 can also occur in clouds, fog, and aerosol liquid water. The reaction rate is highly pH-dependent and it is generally believed that the bisulfide ion (HS^-) is the dominant reactive species in solution. This pathway can be important for in-cloud sulfate production. High pH leads to a faster oxidation rate of reduced sulfur (S(-II)) in cloud water, and consequently more rapid production of sulfate (SO_4^{2-}) in-cloud. Alkaline clouds (e.g., influenced by ammonia or mineral dust) can efficiently convert $\text{H}_2\text{S} / \text{HS}^-$ to sulfate.



NH_3 – Water Equilibrium

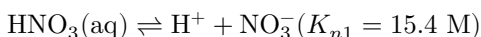


$$[\text{NH}_4^+] = \frac{H_{\text{NH}_3}K_{a1}}{K_w}p_{\text{NH}_3}[\text{H}^+], [\text{NH}_3] = H_{\text{NH}_3}p_{\text{NH}_3} \left(1 + \frac{K_{a1}[\text{H}^+]}{K_w}\right)$$

$$\text{Ammonium fraction: } \frac{[\text{NH}_4^+]}{[\text{NH}_3]} = \frac{K_{a1}[\text{H}^+]}{K_w + K_{a1}[\text{H}^+]}$$

In cloudwater (pH 3–6), NH_3 is essentially completely scavenged from the gas phase. Gas-phase NH_3 concentration inside a cloud is near zero. At low pH (acidic), $K_{a1}[\text{H}^+] \gg K_w \rightarrow$ fraction ≈ 1 . Almost all dissolved ammonia is NH_4^+ (ammonium ion). At high pH (alkaline), $K_{a1}[\text{H}^+]$ becomes small $\rightarrow$ fraction decreases $\rightarrow$ more exists as NH_3 (aq).

HNO_3 – Strong Acid



Because $K_{n1} \gg [\text{H}^+]$ for typical pH, virtually all is NO_3^- :

$$[\text{HNO}_3] \approx \frac{H_{\text{HNO}_3}K_{n1}}{[\text{H}^+]}, p_{\text{HNO}_3} = \frac{3.2 \times 10^6}{[\text{H}^+]}p_{\text{HNO}_3}$$

Aqueous fraction (for cloud L = 0.1–1 g/m³, pH > 2) $\rightarrow$ virtually 1. Hence, HNO_3 is completely scavenged by cloud droplets.

15 Aerosol & Gas Phase Measurement Methods

Monitoring Approaches

Feature	Manual	Automated (Real-Time)
Sampling	Filter or canister, later lab analysis	Continuous in-situ measurement
Time resolution	Hours to days (e.g., 24-hr filter)	Seconds to minutes
Examples	Hi-Vol sampler, passive samplers	TEOM, chemiluminescence analyzers
Pros	Low cost, can archive samples	High time resolution, immediate data
Cons	Labor intensive, no real-time warning	Higher capital cost, power needed

Federal Reference Method (FRM) vs. Equivalent Method (EM)

- FRM: Strictly defined by EPA – must be used for compliance. Any approved method gives legally comparable results.
- EM: Technically different but shown to agree with FRM within limits. Often cheaper or faster.

Criteria Pollutants

Pollutant	Main Sources	Health & Environmental Impacts
SO ₂	Combustion of coal/oil (power plants, industry), ships, volcanoes	Respiratory irritation, aggravates asthma; forms sulfate aerosol (PM _{2.5}) and acid rain.
NO ₂	High-temperature combustion (vehicles, power plants)	Lung inflammation, decreases lung function; precursor to O ₃ , nitrate aerosol, and acid rain.
CO	Incomplete combustion (cars, fires, stoves)	Reduces oxygen delivery to organs (carboxyhemoglobin); headaches, dizziness, fatal at high levels.
O ₃ (ground-level)	Formed from NO _x + VOCs in sunlight (photochemical smog)	Lung damage, chest pain, coughing; reduces crop yields and damages vegetation.
PM ₁₀	Dust, construction, sea salt, combustion, secondary aerosols	Respirable – penetrates upper airways; causes asthma, bronchitis, cardiovascular effects.
PM _{2.5}	Combustion (diesel, wood), industrial emissions, secondary sulfate/nitrate/organics	Deep lung penetration, systemic inflammation, heart attacks, lung cancer, premature death.
Pb	Industrial processes, historical gasoline (still from batteries/smelting)	Neurotoxin – impairs cognitive development in children, cardiovascular and kidney effects in adults.

O₃ is secondary (not directly emitted), while CO, SO₂, NO₂, Pb are primarily primary pollutants. PM can be both primary and secondary.

Gaseous Pollutant Methods

SO₂

● FRM – Spectrophotometry (Pararosaniline)

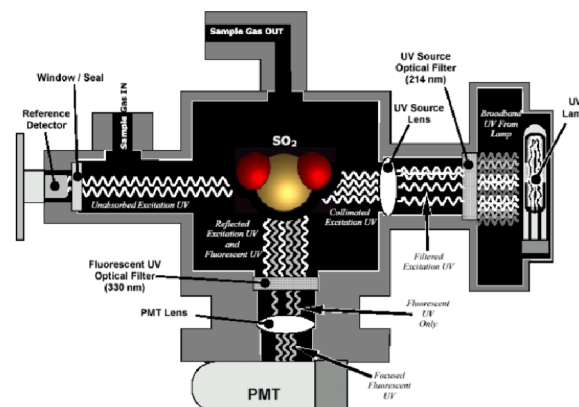
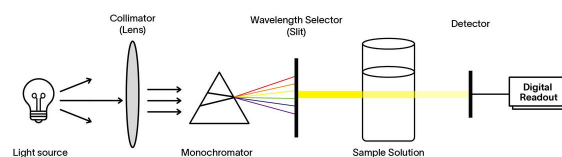
Sample air bubbled through potassium tetrachloromercurate (K₂HgCl₄) solution → forms stable HgCl₂SO₃²⁻ complex. Add formaldehyde (HCHO) and pararosaniline (colorless) → red-violet product. Measure absorbance at 560 nm. Concentration proportional to absorbance.

Issues: interference from NO_x, heavy metals, or ozone can cause color bleaching or enhancement; slow response (not real-time).

● EM – UV Fluorescence

UV lamp (214 nm) → excitation of SO₂ → SO₂* decays to ground state emitting UV (peak ~350 nm) → photomultiplier tube detects fluorescence. A reference cell removes SO₂ to subtract background.

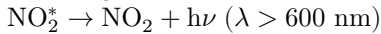
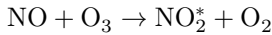
Issues: quenching by water vapor or aromatic hydrocarbons; zero drift requires frequent calibration.



NO/NO₂

- **FRM – Gas-Phase Chemiluminescence**

O₃ generator → O₃ mixed with sample air in a reaction chamber.



Light detected by photomultiplier tube.

For NO₂ measurement: sample first passes through a molybdenum converter (heated to 325 °C) that reduces NO₂ to NO, then measure total NO_x. NO₂ = NO_x – NO. 2NO₂ + Mo → 3NO + MoO₃

Issues: converts other N-compounds (HNO₃, PAN, organic nitrates) → positive bias in NO₂ measurement; high temperature converter can create NO from NH₃ or other sources; interference from water vapor (quenching of chemiluminescence).

- **EM – FT-IR Spectrometry**

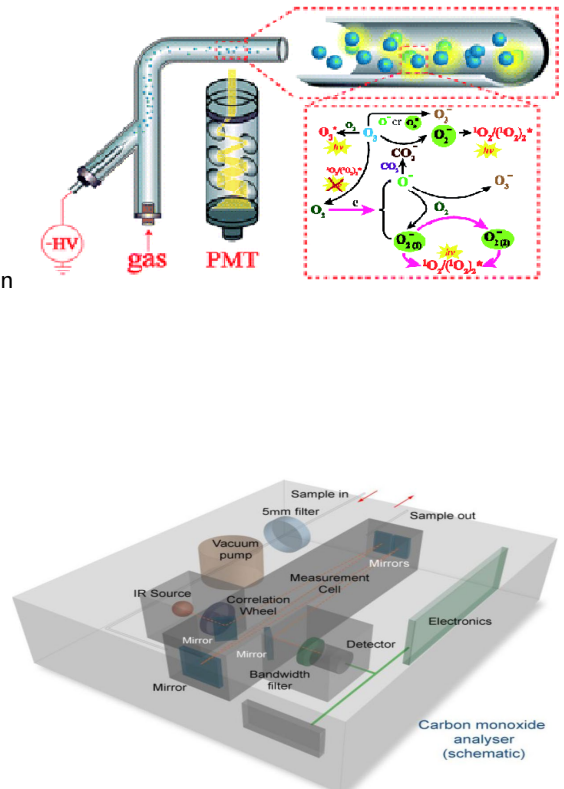
CO

- **FRM – Nondispersive Infrared (NDIR) Spectrometry**

CO strongly absorbs infrared energy at wavelengths near 4.7 microns.

IR source (4.7 μm) passes through two infrared cells – sample cell (sample air) and reference cell (N₂ or CO-free air). CO absorbs IR in sample cell, reducing energy reaching detector. Detector (e.g., gas-filled microphone) senses difference. The greater the absorption, the higher the CO concentration.

Issue: requires gas filter correlation or scrubbers to correct; poor sensitivity at very low concentrations (background ~100 ppb difficult).



O₃

- **FRM – Chemiluminescence**

Light emissions produced on reaction of O₃ with ethylene (C₂H₄)

However, C₂H₄ is flammable – replaced by Rhodamine B dye embedded in a disk, does not attain stable baseline rapidly after exposure to O₃.

- **EM – UV Photometry**

Sample air passes through a scrubber (MnO₂ or heated catalyst) that removes O₃ from one stream (reference). Both sample and reference streams alternate through a 254 nm UV lamp. Detector measures intensity difference. O₃ concentration → Beer-Lambert law.

Interference: Aromatic hydrocarbons (e.g., xylenes) also absorb at 254 nm → positive bias. Some monitors use a selective scrubber that removes O₃ but not VOCs.

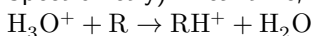
NMHCs (Non-Methane Hydrocarbons)

- **FRM – GC-FID**

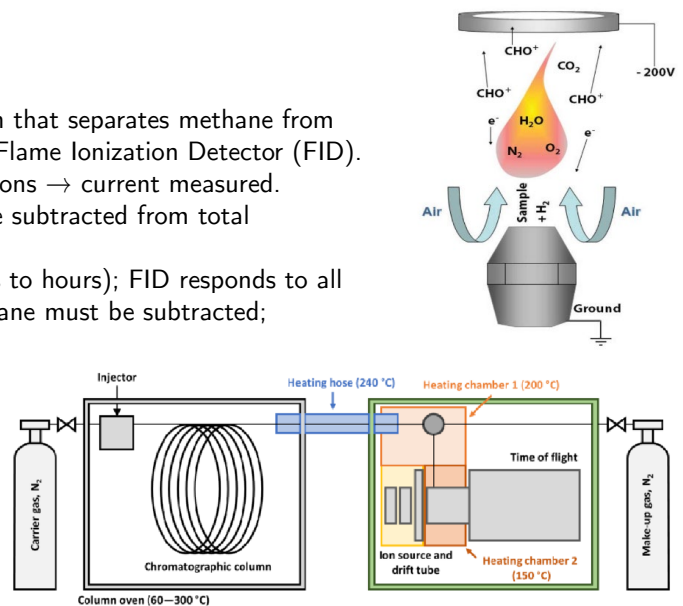
Sample air drawn into a gas chromatograph with a column that separates methane from other hydrocarbons. After separation, compounds enter a Flame Ionization Detector (FID). Hydrogen flame combusts hydrocarbons → ions and electrons → current measured. Methane passes first, then heavier NMHCs. Methane value subtracted from total hydrocarbons to get NMHCs.

Issues: not real time (sampling + separation takes minutes to hours); FID responds to all hydrocarbons (cannot speciate without GC column); methane must be subtracted; incomplete separation possible.

- **PTR-MS (Proton Transfer Reaction Mass Spectrometry) – real-time, no column.**



RH⁺ is detected by mass spectrometer. Very fast (seconds), but expensive, requires careful calibration, and cannot separate isomers.

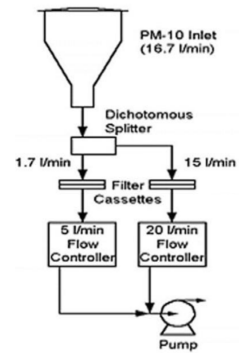


Particulate Matter (PM) Methods

● PM₁₀ & PM_{2.5} Size-Selective Inlets Component

Principle (impactor): Air accelerated through a nozzle toward a flat plate. Large particles (> cutoff) have inertia → hit plate and stick. Small particles follow streamlines around plate. Cutoff diameter depends on nozzle diameter, flow rate, and plate distance.

EM – Virtual impactor (dichotomous sampler): No solid plate – large particles go into a small “collection” stream (1/10 of total flow), small particles go into main stream. It is a two-stage impactor dividing the particles into two size ranges, 10 – 2.5 µm and less than 2.5 µm.



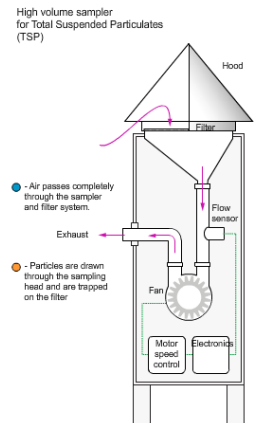
● High-Volume Sampler (Hi-Vol)

Large blower draws air through a filter (typically 8"×10"). Particles collected on filter. Weigh filter before and after (after equilibration in controlled humidity/temperature). Known air volume (e.g., 1–2 m³/min over 24 h) → mass concentration (µg/m³).

If no size-selective inlet is attached, all particles up to ~100 µm (TSP, Total Suspended Particulate) are collected.

By impaction on a greased surface, particles >10 µm can be removed and particles <10 µm can be collected on a quartz glass fiber filter.

Issue: not real-time, semi-volatiles can evaporate during sampling.

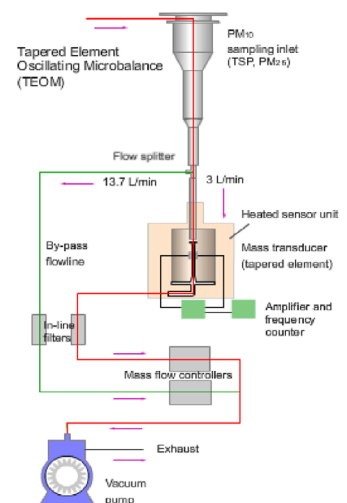


● TEOM (Tapered Element Oscillating Microbalance)

Real-time instrument (EM for PM₁₀, PM_{2.5}). A hollow tapered tube (like a tuning fork) with a filter at its tip. Air flows through filter → particles deposit. Tube oscillates at its natural frequency.

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

As particle mass (m) increases, frequency (f) decreases. A microprocessor measures frequency change every few seconds to infer real-time mass concentration. Instrument heated to 50 °C to avoid water condensation, but this can cause semi-volatiles (e.g., NH₄NO₃) to evaporate, as a negative bias compared to ambient PM.



● Cascade Impactors (e.g., MOUDI – Micro-Orifice Uniform Deposit Impactor)

Non-real time device that measures size-resolved mass distribution.

Small particles, because of their small inertia, can follow the gas streamlines and are not deposited.

Large particles, due to their higher inertia, cannot follow the streamlines and are deposited on the impaction plate.

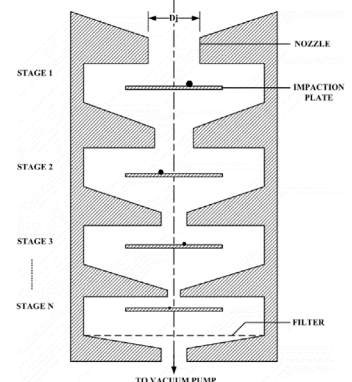
Multiple stages stacked, each with smaller nozzles than previous.

Air passes through stage 1 (largest cutoff) → coarse particles collected.

Then through stage 2 (smaller cutoff) → medium particles, etc.

Last stage has a backup filter for finest particles.

After sampling, each stage is weighed or analyzed chemically.



Particle Size & Number Instruments

- **DMA (Differential Mobility Analyzer)**

Aerosol first passes through a **neutralizer** (e.g., ^{85}Kr or X-ray) to give particles a known charge distribution (mostly +1, -1, or 0). Then into DMA column: inner rod at high voltage (positive or negative), outer cylinder grounded. Sheath air (clean, particle-free) flows downward. Charged particles are attracted to the rod. Only particles with a specific electrical mobility exit through a slit at the bottom – others hit rod or go to exhaust. By scanning voltage, one can get a size distribution of mobility diameter.

- **CPC (Condensation Particle Counter)**

Particles from DMA enter a heated saturator filled with butanol. Butanol vapor condenses on particles in a cooled condenser → particles grow to $\sim 10\ \mu\text{m}$ droplets. Droplets pass through a laser beam → each droplet scatters light → photodetector counts pulses → number concentration ($\#/\text{cm}^3$). CPC can detect particles as small as 2.5 nm (though DMA alone classifies down to $\sim 10\ \text{nm}$).

- **Combined DMA+CPC** → scan voltage → measure number concentration at each size → plot $dN/d\log D_p$ vs. D_p .

- **APS (Aerodynamic Particle Sizer)**

Particles are accelerated through a nozzle. A double-laser beam system (two parallel laser sheets) measures **time-of-flight** between the two beams. Larger aerodynamic diameter → lower velocity → longer flight time. Calibration with known particles (e.g., PSL spheres) gives size. Measures from ~ 0.5 to $20\ \mu\text{m}$. Good for coarse and accumulation modes.

- **Optical Particle Counter (OPC)**

Single particles pass through a focused laser beam. Light scattered at $\sim 90^\circ$ is collected and converted to a voltage pulse. Pulse height relates to **optical diameter** (function of refractive index). Counts particles in preset size bins. Fast but less accurate for non-spherical or absorbing particles.

Chemical Composition – Real-Time

- **ACSM / AMS (Aerosol Chemical Speciation Monitor / Aerosol Mass Spectrometer)**

1. Inlet: Aerosol sampled through an aerodynamic lens → forms a narrow particle beam.
2. Size selection: Optional – a chopper or particle time-of-flight chamber (AMS only) measures vacuum aerodynamic diameter.
3. Flash vaporization: Particles hit a heated surface ($\sim 600^\circ\text{C}$) – non-refractory components (sulfate, nitrate, ammonium, organics, chloride) instantly vaporize. Refractory material (sea salt, dust, black carbon) mostly passes through or is collected separately.
4. Electron impact ionization: Vapor molecules hit 70 eV electrons → ionized.
5. Mass spectrometer (quadrupole or time-of-flight) detects ions by m/z .
6. Data: Real-time concentrations ($\mu\text{g}/\text{m}^3$) of major components. Can also separate organics into e.g., hydrocarbon-like, oxygenated, cooking-related using factor analysis.

Key feature: No filter collection, no lab extraction – fast (minutes) time resolution. Measures non-refractory PM₁ (particles $< 1\ \mu\text{m}$).

Limitation: Does not measure black carbon, dust, metals, or NaCl (sea salt is partly refractory – some signal but undercounted).